Next.js 15 Production & Architecture Toolkit

Zero-Bailout Static Generation, Bilingual Routing, Security Headers & CI Verification

1. App Router Architecture & Zero-Bailout SSG

Next.js 15 App Router defaults to Server Components, but subtle client hooks can silently force an entire route to bail out to client-side rendering (CSR). Follow these architectural rules to keep 100% of your marketing and case study HTML pre-rendered at build time.

- [ ] **Avoid Top-Level useSearchParams() in Shared Chrome**
  Calling useSearchParams() inside a Header, LanguageToggle, or TagFilter without a <Suspense> boundary causes the entire page to emit BAILOUT_TO_CLIENT_SIDE_RENDERING. Read window.location.search inside click/change handlers instead.
- [ ] **Eliminate Unnecessary Root loading.tsx Boundaries on Static Routes**
  A root app/loading.tsx wraps static pages in a streaming Suspense boundary (<div hidden id="S:0">), hiding content from non-JS scrapers and delaying initial DOM visibility.
- [ ] **Never Render opacity: 0 as the Initial SSR State**
  Scroll-reveal and text-split animations that start at opacity: 0 render invisible text in SSR HTML, hurting Lighthouse LCP and causing blank captures in automated preview tools.

```
// GOOD: Event-time URL query preservation without CSR bailout
const handleLocaleChange = (nextLocale: Locale) => {
  const query = typeof window !== "undefined" ? window.location.search : "";
  router.push(‘/${nextLocale}${cleanPath}${query}‘);
};
```

Verification Rule: Run curl -s https://your-domain.com/en | grep -c "BAILOUT_TO_CLIENT_SIDE_RENDERING" after every build to confirm your primary content renders in static HTML.

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2. Bilingual i18n, SEO Metadata & Crawler Access

- [ ] **Static Locale Pre-Rendering via generateStaticParams**
  Export generateStaticParams() in app/[locale]/layout.tsx so Next.js pre-builds /en and /fr routes as static HTML files at build time.
- [ ] **Complete Hreflang & Canonical Alternates**
  In generateMetadata(), populate alternates.canonical and alternates.languages for every supported locale plus x-default so search engines index localized variants accurately.
- [ ] **Never Disallow /_next/ in robots.ts**
  Blocking /_next/ in robots.txt prevents Googlebot and design/SEO auditors from loading CSS stylesheets, web fonts, and optimized images, causing pages to render as unstyled text.
- [ ] **Server-Level Root Redirect in next.config.ts**
  Configure a server redirect from / to /en (temporary 307) alongside middleware locale negotiation so HTTP clients without cookies land cleanly on the default locale.

3. API Route Validation, Rate Limiting & Security Headers

```
// API Route Defense-in-Depth Pattern (Zod + Upstash Redis)
const ratelimit = new Ratelimit({
  redis: Redis.fromEnv(),
  limiter: Ratelimit.slidingWindow(5, "1 m"),
});
const parsed = ContactSchema.safeParse(await req.json());
if (!parsed.success) return NextResponse.json({ errors: parsed.error.flatten() }, { status: 400 });
```

- [ ] **Strict HTTP Security Headers**
  Configure Content-Security-Policy (CSP), Strict-Transport-Security (HSTS), X-Frame-Options: DENY, X-Content-Type-Options: nosniff, and Permissions-Policy in middleware or next.config.ts.

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4. Production Launch & CI/CD Verification Checklist

- [ ] **Type Safety & Content Schema Compilation**
  Run contentlayer2 build && tsc --noEmit in CI to validate MDX frontmatter schemas and TypeScript types before building.
- [ ] **Unit & Integration Suite (Vitest)**
  Test Zod validation schemas, i18n routing helpers, email client error handling, and component ARIA contracts in Vitest.
- [ ] **End-to-End Accessibility & Flow Verification (Playwright)**
  Pin runs-on: ubuntu-24.04 in GitHub Actions CI. Verify html[lang], skip-to-content links, keyboard Tab navigation, locale switching with query preservation, and contact wizard submission.
- [ ] **Real-Time Analytics & Transactional Email**
  Verify SPF/DKIM records on your custom domain for Resend delivery, and wire privacy-friendly event tracking (Plausible / Vercel Analytics) on primary CTAs.

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